

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			Ticks in 1 st , 2 nd and 3 rd boxes $3 \times (1)$ 4 boxes ticked maximum = 2 marks 5 boxes ticked maximum = 1 mark 6 boxes ticked = 0 marks	3			3		
	(b)			Beta emitters are most suitable as beta would be partly absorbed [Accept: beta will pass through <u>thin</u> Al / blocked by a few mm [or thick] Al] / alpha totally absorbed and gamma not absorbed (1) Increasing thickness decreases beta <u>count rate</u> [accept: changing thickness would change count rate] (1) Sr-90 has a long enough half-life so won't need frequent replacing / P-32 would need frequent replacing. (1) For 3 marks "agree" required.			3	3		
	(c)	(i)		It takes 29 years / this is the time <u>to halve</u> (1) number of nuclei / atoms / mass / amount / activity / count rate [of strontium-90] (1)	2			2		
		(ii)		[1 $\rightarrow$] $\frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8}$ or [100] $\rightarrow 50 \rightarrow 25 \rightarrow 12.5\%$ (1) multiple halving ending with $\frac{1}{8}$ (12.5%) so $\frac{1}{8}$ is 3 half-lives ecf on incorrect halving or incorrect counting of half-lives (1) 3 ecf $\times 29 = 87$ years (1) NB 87 years $\rightarrow$ 3 marks; 58 or 116 years $\rightarrow$ 2 marks		3		3	3	